



8.13 Infectious Neck Masses in Children

◆ Key Features

- Cervical lymphadenitis is the most common infectious neck mass in children, but an abscess is also a possibility.
- The age of the patient, the location, and the time frame to development of adenitis are important clues to the underlying etiology.
- The diagnosis is commonly based on history and physical exam, not laboratory findings.
- Medical management is appropriate for many causes.

Enlarged or inflamed lymph nodes are the cause of 95% of pediatric neck masses. A thorough physical examination is necessary because the lymph node groups are associated with different diseases, which will dictate proper management. Most of the lymphatics from the head and neck region drain to the submaxillary and deep cervical lymph nodes, explaining why these nodes are most commonly affected by cervical lymphadenitis. In the absence of cervical adenopathy, enlargement of the supraclavicular nodes can be indicative of thoracic or abdominal disease. The most common causes of isolated right supraclavicular node enlargement are Hodgkin's and non-Hodgkin's lymphoma. Isolated enlargement of the left supraclavicular

nodes is most commonly associated with intraabdominal tumor or inflammation (Troisier sign).

◆ Etiology

The underlying etiology of cervical lymphadenitis can be predicted based on the patient's age. In neonates, group B streptococci are the most common cause of lymphadenitis. *Staphylococcus aureus* is usually the causative organism in patients 2 months to 1 year old. Along with *Bartonella henselae* (cat-scratch disease) and nontuberculous mycobacteria, *S. aureus* is also a common cause in 1- to 4-year-old patients. These organisms can be the cause of cervical adenitis in older patients along with tuberculosis, anaerobic bacteria, and toxoplasmosis. The timeframe is also an important factor to consider when determining the etiology of cervical adenitis. Acute bilateral disease is usually a response to acute pharyngitis but can also occur with Epstein-Barr virus, cytomegalovirus, herpes simplex virus, roseola, and enteroviruses. Acute unilateral lymphadenitis commonly presents with an associated cellulitis and is typically caused by *S. aureus*, *Streptococcus pyogenes*, and group B streptococci. Subacute or chronic unilateral lymphadenitis is much less common. It can be due to cat-scratch disease, mycobacterial infection, or toxoplasmosis.

◆ Clinical

Signs

A thorough physical examination is essential. Note the size, location, laterality, firmness, and number of nodes. Erythema and tenderness are also important signs. Associated illnesses such as pharyngitis or systemic infection need to be noted.

Symptoms

Patients can present with unilateral or bilateral neck swelling, with or without any other symptoms. Tenderness of the affected nodes is commonly associated with acute infection.

Differential Diagnosis

Infectious causes are the most common. Malignancy is an important consideration in patients with no other signs of infection, with recent weight loss or fevers, or with isolated supraclavicular involvement. Noninfectious causes are much less common and include Kawasaki's disease, sarcoidosis, sinus histiocytosis, histiocytic necrotizing lymphadenitis, and Kimura's disease.

◆ Evaluation

Physical Exam

Cervical lymphadenitis typically presents with acute unilaterally or bilaterally enlarged (> 3 cm) and tender lymph nodes in the jugulodigastric area. Atypical mycobacterial infection generally presents with an enlarged,

erythematous single mass, distinct from reactive adenopathy or fluctuant abscess.

Imaging

Imaging is not necessary if cervical lymphadenitis is suspected based on history and physical exam findings. If there is concern for tuberculosis or an abscess, an ultrasound or computed tomography (CT) of the neck can be performed. Cervical lymphadenitis will manifest as enlarged, enhancing nodes with low central attenuation if necrosis is present. Tuberculosis-infected lymph nodes will have thick peripheral enhancement on contrasted CT, low central attenuation, a relative lack of fat stranding, and calcification.

Labs

Complete blood count (CBC) is useful for determining whether the enlarged lymph nodes are secondary to systemic infection. Other laboratory studies, including Gram staining, acid-fast staining, and culture, can be done if aspiration is undertaken. Perform cat-scratch titer or monospot if infection is suspected.

Other Tests

Fine-needle aspiration biopsy (FNAB) can be both diagnostic and therapeutic. Obtain the aspirate from the largest, most fluctuant node using a 23- or 20-gauge needle. Gram stain, acid-fast stain, and culture should be obtained. The etiology is discovered in 60 to 90% of patients who undergo needle aspiration. Excisional biopsy is indicated if the node is hard, is fixed, fails to regress following aspiration or antibiotic use, enlarges, or is associated with fever or weight loss or if the diagnosis is uncertain. Biopsy of more than one node is preferable. Placing a portion of the specimen in a flow cytometry medium is important if lymphoma evaluation is required.

Pathology

Pathology varies depending on the underlying etiology.

◆ Treatment Options

Medical

Patients who have asymptomatic, small (< 3 cm), bilaterally enlarged cervical lymph nodes can be observed. If mass persists or enlarges over several weeks, biopsy is warranted. Those with signs and symptoms typical of acute bacterial lymphadenitis (large, tender, erythematous unilateral node with no systemic symptoms) can be treated empirically for *S. aureus* and *S. pyogenes* with amoxicillin-clavulanate, cephalexin, or clindamycin. If cellulitis is present or if the patient is having severe symptoms, parenteral nafcillin, cefazolin, or clindamycin is appropriate. When lymphadenitis is secondary to dental infection, anaerobic infection should be suspected, and clindamycin or penicillin plus metronidazole is effective. Azithromycin, trimethoprim-sulfamethoxazole, or rifampin is effective early in the course of disease in preventing abscess formation if cat-scratch disease is suspected.

Relevant Pharmacology

- Amoxicillin-clavulanate: Amoxicillin binds to penicillin-binding protein, thereby preventing bacterial cell wall synthesis. Clavulanate inhibits β -lactamases, which increases amoxicillin's spectrum of activity. Dosing is based on the amoxicillin component.
 - < 3 months old: Total dose is 30 mg/kg/day. Divide dose and administer twice daily using the 125 mg/5 mL suspension. Children < 40 kg: 25 to 45 mg/kg/day divided every 12 hours using either 200 mg/5 mL or 400 mg/5 mL suspension. Alternatively, 200- or 400-mg chewable tablets can be used. Children $\geq$ 40 kg and adults, 875 mg twice daily.
- Clindamycin: This antibiotic inhibits bacterial protein synthesis by binding to bacterial 50S ribosomal subunits.
 - Children < 16 years
 - Oral: total dose of 8 to 25 mg/kg/day in 3 or 4 divided doses
 - Parenteral: 15 to 20 mg/kg/day
 - Adults
 - Oral: 300 mg three times daily
 - Parenteral: 1.2 to 1.8 g/day in two to four divided doses
- Trimethoprim-sulfamethoxazole: Both agents inhibit bacterial folic acid production. Trimethoprim inhibits dihydrofolic acid reduction, and sulfamethoxazole interferes with dihydrofolic acid. Dosing is based on trimethoprim component.
 - Children > 2 months
 - Total dose: Trimethoprim 8 to 12 mg/kg/day in divided doses every 12 hours or 20 mg/kg/day in divided doses every 6 hours for serious infections
 - Children > 40 kg or adults
 - Oral: Trimethoprim 160 mg every 12 hours
 - Parenteral: Trimethoprim 2 mg/kg every 6 hours

Surgical

Incision and drainage is useful if there is an abscess, especially due to *S. aureus* or *S. pyogenes*. If a mycobacterial or *Bartonella henselae* infection is suspected, then FNAB is preferable to avoid fistula formation. Most cases of cat-scratch fever are managed medically. Incision and drainage are reserved for cases progressing to abscess formation. Surgical excision or curettage is effective if nontuberculous mycobacterial infection is the cause. Removal of the largest node and necrotic nodes is sufficient, because the remaining adenopathy will resolve spontaneously.

◆ Complications

Infection control is essential if a postoperative complication occurs. Empiric use of broad-spectrum antibiotics is appropriate. Further care depends on the type of complication. Internal jugular venous thrombosis can be managed with anticoagulation. Abscess is treated with incision and drainage.

◆ Outcome and Follow-Up

The patient can return home after incision and drainage or surgical excision, but close monitoring is necessary because of the potential complications. Mediastinal abscesses, purulent pericarditis, thrombosis of the internal jugular vein, pulmonary emboli, or mycotic emboli are all rare, but serious, complications.

Cervical lymphadenitis resolves completely in the majority of patients who receive appropriate antibiotic therapy. No further follow-up is necessary unless the patient's symptoms do not improve or if they worsen.

8.14 Hemangiomas, Vascular Malformations, and Lymphatic Malformations of the Head and Neck

◆ Key Features

- Hemangiomas, vascular malformations, and lymphatic malformations of the head and neck are common congenital and neonatal abnormalities.
- Vascular tumors are classified into hemangiomas and vascular malformations.
- Accurate diagnosis is the key to prognosis and treatment plan.
- Accurate diagnosis is based on natural history and key clinical features.

Vascular tumors are common pediatric anomalies and may develop throughout the head and neck. They are divided broadly into hemangiomas and vascular malformations, each with a distinct natural history. Lymphatic malformations (or lymphangiomas) are defects in the lymphatic system that present as swellings or nodules in the skin or mucous membranes of the head and neck. The malformation of lymphatic tissue causes an accumulation of fluid, which accounts for the clinical presentation.

◆ Epidemiology

Infantile hemangiomas occur in up to 12% of all children. They are more common in females than males, at a ratio of 3:1, and more common in European Americans. Sixty percent of hemangiomas are located in the head and neck, including the upper aerodigestive tract, and 80% occur as single lesions.

Vascular malformations are divided into lymphatic malformations, capillary malformations or portwine stains, venous malformations, and high-flow arteriovenous malformations (AVMs). Lymphatic malformations are further divided into microcystic lymphatic malformations and macrocystic lymphatic malformations involving the soft tissue of the neck. Vascular malformations have no gender or racial predilection and most commonly occur in the head and neck. Lymphatic malformations represent 6% of

benign cervicofacial tumors of childhood; they are seen in fewer than 2.8 per 1000 people. Patients with Down's, Turner's, or fetal alcohol syndrome have a higher incidence of lymphatic malformations.

◆ Clinical

Signs and Symptoms

Hemangiomas

Infantile hemangiomas are generally not present at birth and first appear during the first 6 weeks of life. Hemangiomas go through predictable phases of growth. The initial proliferative phase occurs over the first year, is characterized by rapid growth, and is followed by the involution phase with subsequent regression. Complete involution occurs in 50% of children by 5 years of age, 70% of children by 7 years of age, and 90% of children by 9 years of age.

Vascular Malformations

Vascular malformations are by definition present at birth but may go unrecognized. They grow proportionately with the child and may present throughout childhood or early adulthood. Hemangiomas are firm; vascular malformations are easily compressible. They may involve the facial skeleton and abruptly increase in size with acute hemorrhage or infection, puberty, or pregnancy.

Lymphatic Malformations

Lymphangioma circumscriptum presents as a nodular mass with a red, wart-like appearance. They are typically asymptomatic, but bleeding or drainage of fluid may occur.

Cavernous lymphangioma (cystic hygroma) presents as a painless subcutaneous swelling or nodule and can rapidly enlarge with infections.

Differential Diagnosis

The key to diagnosis is to differentiate between hemangiomas and vascular malformations. Diagnosis can generally be made on the clinical history and physical exam alone. Lymphatic malformations presenting as cystic lesions within the neck must be differentiated from other congenital cystic lesions, including branchial cleft cysts (discussed in **Chapter 8.15**).

◆ Evaluation

History

In the patient history, the timing of the development of the lesion is important, particularly whether it was present at birth. Also note the growth rate of the lesion.

Physical Exam

A complete head and neck exam should focus on the character of the lesion. Children with hemangiomas in the chin and chest are more likely to

have concurrent airway lesions, including subglottic hemangiomas. Noisy breathing in these children must be evaluated with laryngoscopy and likely with bronchoscopy.

Imaging

Imaging is generally unnecessary for diagnostic purposes but may be warranted in children with multiple cutaneous hemangiomas, who are more likely to have concurrent visceral lesions. Magnetic resonance imaging (MRI) may be useful, particularly in the preoperative setting, to determine the extent of soft tissue involvement. Imaging may also be helpful in the evaluation of a child with cutaneous hemangioma and noisy breathing. Flow voids are indicative of high-flow lesions.

Pathology

Pathologic exam also helps to differentiate between hemangioma and vascular malformation. Hemangiomas are characterized by proliferating endothelial cells and are positive for GLUT1, helping to differentiate them from other vascular tumors. Hemangiomas grow by cellular hyperplasia; vascular malformations grow by hypertrophy.

◆ Treatment Options

Hemangiomas

Expectant management is recommended for most hemangiomas. Intervention is indicated for lesions threatening the airway or vision and may be considered for bleeding and infection or progressive disfigurement. Tracheotomy is sometimes required to manage the airway of a child with a subglottic hemangioma.

The first-line therapy for hemangiomas is propranolol, a nonselective β -blocker. Side effects or lack of response to propranolol therapy are rare, but selective β -blockers may be used if side effects are noted due to the nonselective nature of propranolol. Topical β -blocker therapy with timolol ophthalmic gel is commonly used for superficial, localized hemangiomas.

Corticosteroids were previously used as first-line therapy for hemangiomas. However, the short- and long-term complications associated with corticosteroid therapy have limited their use. Corticosteroids are often reserved for patients with contraindications to β -blocker therapy or as an acute therapy for patients being started on β -blocker therapy.

Interferon has been used with success but is associated with neurologic complications. The use of various lasers may be considered. Surgery may be appropriate, but must be considered carefully.

Vascular Malformations

Vascular malformations are not expected to involute, and so the decision to treat is less controversial. Tracheotomy may be required for large cystic hygromas. Sclerotherapy with OK-432 (picibanil) has shown success, though OK-432 remains an experimental treatment modality. There are other older sclerotherapy agents used widely with success, such as ethanol.

Argon lasers and dye lasers are used in the treatment of capillary malformations or portwine stains (see **Chapter 6.3.14**). Complete surgical resection, with the consideration of preoperative embolization, is required in the treatment of AVMs.

Lymphatic Malformations

Surgical management of lymphatic malformations may be considered, although the lesion may involve vital structures and complete resection may not be appropriate.

◆ Complications

Complications of enlarging vascular tumors and surgical intervention depend on the size and location of the lesion. Functional and cosmetic factors must be considered in both the decision to treat and the extent of treatment.

◆ Outcome and Follow-Up

Hemangiomas initially managed expectantly must be monitored closely, as even lesions not involving the airway or orbit may cause significant stress to parents during the proliferative phase. Untreated vascular malformations must be similarly monitored as they grow proportionately with the child. Large lymphatic malformations can require multiple surgeries, require tracheotomy, and cause significant morbidity.

8.15 Branchial Cleft Cysts

◆ Key Features

- Branchial cleft cysts represent the most common noninflammatory lateral neck masses in children.
- Definitive treatment is complete surgical excision.

Branchial cleft cysts are composed of remnants from any of the first five branchial arches, which give rise to head and neck structures (**Table 8.10**). Branchial cleft cysts may be associated with a fistula or sinus tract. Cysts may have different locations and characteristics depending on their branchial cleft of origin.

◆ Epidemiology

Branchial cleft cysts are common; they are responsible for 33% of congenital neck masses and 17% of pediatric neck masses. Most branchial cleft cysts are derived from the second branchial arch apparatus; 1% of cysts

are derived from the first branchial arch. Third-arch cysts are reported; fourth-branchial-cleft cysts are extremely rare.

Table 8.10 Structures that arise from differentiation of branchial arches

Arch	Nerve	Structure	Muscle	Artery
First	Trigeminal	Mandible, body of incus, head and neck of malleus, major salivary glands, tympanic membrane, eustachian tube	Masticator muscles, tensor tympani, anterior belly of digastric	Facial
Second	Facial (CN VII), vestibulocochlear (CN VIII)	Two-thirds of long process of incus, manubrium of malleus, crura and head of stapes, styloid, lesser cornu of hyoid, upper body of hyoid, tonsil	Platysma, muscles of facial expression, stapedius, posterior belly of digastric	Stapedial
Third	Glossopharyngeal (CN IX)	Greater cornu of hyoid, thymus, inferior parathyroid, body of hyoid	Superior constrictors	Internal carotid
Fourth	Vagus (CN X)	Thyroid cartilage, epiglottis, superior parathyroid	Inferior constrictor, laryngeal musculature	Aortic arch, right subclavian
Fifth	Spinal accessory (CN XI)	Arytenoid cartilage, cricoid cartilage, lungs	Portion of laryngeal musculature	Pulmonary ductus

Used with permission from Van de Water TR, Staecker H. *Otolaryngology: Basic Science and Clinical Review*. Stuttgart/New York: Thieme;2006:208. Abbreviation: CN, cranial nerve.

◆ **Classification**

First branchial arch abnormalities can be classified as type 1 or 2:

- Type 1 abnormalities are of ectodermal origin and are lined by epidermoid elements. They lie parallel to the external auditory canal in the preauricular area. They may be associated with a fistula or sinus tract that may end in the middle ear or external auditory canal.

- Type 2 abnormalities are more common than type 1. They are of ectodermal and mesodermal origin and consist of squamous epithelium and adnexal structures. An external opening may be present, leading to a cyst or sinus tract that courses through the parotid gland. Patients often present with otorrhea unresponsive to typical treatment. Note that this tract may be closely associated with the facial nerve at the stylomastoid foramen.

Second-branchial-arch remnants, the most common branchial anomaly, present as a painless mass or dimple below the angle of the mandible, at the anterior border of the sternocleidomastoid muscle. They may present as a cyst or may have a tract ending at the tonsillar fossa. The tract, if present, traverses the carotid bifurcation (**Fig. 8.4**).

Third- and (especially) fourth- and fifth-branchial-cleft remnants are much more rare; they may have a long tract looping deep to the carotid artery and ending in the region of the piriform.

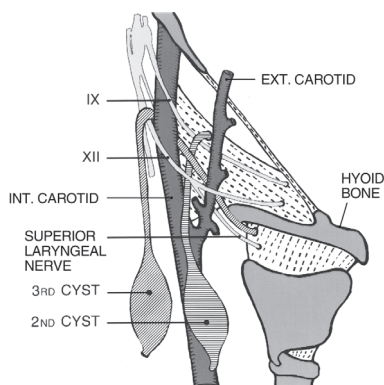


Fig. 8.4 Relationship of second- and third-branchial-cleft anomalies to the carotid artery and cranial nerves. (Used with permission from Van de Water TR, Staecker H. *Otolaryngology: Basic Science and Clinical Review*. Stuttgart/New York: Thieme; 2006:209.)

◆ Clinical

Signs and Symptoms

Branchial cleft cysts often present as lateral painless masses and may enlarge after an upper respiratory infection. If there is an external sinus or fistula opening, there are troublesome mucus secretions; recurrent infections are common. Larger branchial cysts may compress the airway causing stridor, dyspnea, or dysphagia.

Differential Diagnosis

- Branchial cleft cyst or sinus
- Lymphatic malformation
- Hemangioma
- Ectopic thymus

- Lipoma
- Thymic cyst
- Parotid cyst
- Pseudotumor of infancy
- Malignant tumor
- Laryngocele

◆ Evaluation

History

The timing of the development of the lesion is important, particularly whether it was present at birth. Also, note the growth rate, size fluctuation, and history of infection or drainage.

Physical Exam

During the head and neck exam, note the size, the precise location of the mass, the presence or absence of a dimple or opening, and quality (firm, soft, etc.). Assess for possible other congenital anomalies or syndromic features.

Imaging

Branchial abnormalities are visualized using computed tomography (CT) with contrast; if a sinus or fistula is present, contrast fistulography and barium swallow esophagography may demonstrate the path of the abnormality.

Pathology

Branchial cleft cysts arise from residual embryonic tissue from the branchial clefts or pouches (**Table 8.10**).

◆ Treatment Options

Medical

If the branchial cyst is infected, treatment with intravenous (IV) antibiotics is indicated before surgical excision.

Surgical

The treatment of choice for branchial cleft cysts is surgical excision. Excision of a Type 1 first-branchial-cleft cyst may include the cartilage of the external auditory canal. For Type 2 first arch, total parotidectomy with facial nerve dissection may also be needed.

Second-, third-, fourth-, and fifth-branchial-cleft cysts are excised through horizontal neck incisions. “Stepladder” incisions may be needed to excise a tract. The relationship of the tract to carotid and cranial nerves is predictable and must be understood. If a lesion is a true third-branchial cyst, it may present as a thyroid cyst, in which case hemithyroidectomy is performed.

Other treatments such as radiation or sclerosing agents do not provide a cure for branchial cysts and increase the risk of recurrence.

◆ Complications

Bleeding or hematoma can be prevented using electrocautery on small blood vessels and silk ties on larger vessels. Recurrence of branchial cleft cysts can occur if there is incomplete excision of the mass. Injury to the facial nerve may occur during the excision of a first-branchial-cleft cyst. Damage to associated neurovascular structures may occur during the excision of a second- or third-branchial abnormality.

◆ Outcome and Follow-Up

If facial nerve dissection was performed, then assess function in the early postoperative period. A Penrose or suction drain is recommended for any extensive dissection. Antibiotics are necessary if the aerodigestive tract was entered or if the cyst was infected. Postoperative care is otherwise routine.

8.16 Congenital Midline Neck Masses

◆ Key Features

- Thyroglossal duct cysts, dermoid cysts, and lipomas are common midline congenital masses.
- Thyroid cancer can develop from a thyroglossal duct cyst, rarely.
- Treatment requires complete surgical excision.

Thyroglossal duct cysts are midline masses that may appear anywhere between the base of the tongue and the thyroid gland. Dermoid cysts typically present in the submental area of the neck but can appear in other areas of the head and neck. Lipomas are derived from adipose tissue and present as soft painless masses that may occur anywhere in the body.

◆ Epidemiology

Thyroglossal duct cysts are the most common congenital neck mass, present in 7% of the population. Lipomas are the most common soft tissue tumor. They may occur at any age but are common after age 40.

◆ Clinical

Signs and Symptoms

Thyroglossal duct cysts are firm, mobile typically midline masses and can appear anywhere along the path of the thyroid's descent. They can be off-midline. They will elevate upon protrusion of the tongue or swallowing and are usually asymptomatic. However, large cysts, or those in the base of the tongue, can compress the airway, causing dysphagia and respiratory difficulties. Cysts can become infected and may also enlarge after an upper respiratory infection.

Dermoid cysts are attached to and move with the skin. They are usually located in the submental area. They do not move upon swallowing or protrusion of the tongue. They are typically asymptomatic unless infected.

Lipoma presents as a soft, movable lump under the skin. They occur in the neck but also in multiple tissues and locations throughout the body.

Differential Diagnosis

- Thyroglossal duct cyst
- Dermoid cyst
- Lipoma
- Enlarged thyroid isthmus
- Pyramidal lobe of thyroid
- Plunging ranula
- Thyroid nodule/cyst
- Sebaceous cyst
- Teratoma
- Cervical thymic cyst

◆ Evaluation

Physical Exam

During the full head and neck exam, note the location, the quality of the mass, its size, and its motion with skin or swallowing. Assess for possible other congenital anomalies.

Imaging

Ultrasound of the thyroid is mandatory to document thyroid tissue in its normal location, as a thyroglossal duct cyst may rarely contain all functioning thyroid tissue. Ultrasound may be sufficient to diagnose a thyroglossal duct cyst and differentiate it from ectopic thyroid. If the diagnosis is unclear, computed tomography (CT) or magnetic resonance imaging (MRI) will usually differentiate thyroglossal duct cyst, dermoid, and lipoma.

Pathology

Thyroglossal duct cysts arise from remnants of the thyroid gland as it descends from the foramen cecum during embryologic development (**Fig. 8.5**). Dermoid cysts are epithelial-lined cavities containing adnexal structures. They result from epithelium becoming enclosed in tissue during embryological development. Lipomas are benign, subcutaneous masses derived from adipose tissue.

◆ Treatment Options

Medical

If the cyst is infected, treatment with intravenous (IV) antibiotics is indicated before surgical excision.

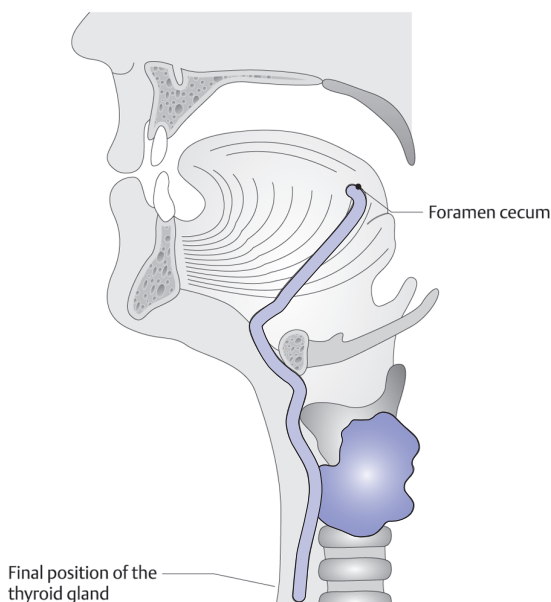


Fig. 8.5 Drawing of the lateral neck showing the path of descent of the thyroid anlage. Thyroid duct cysts, fistulae, and ectopic thyroid tissue can occur anywhere along the course of this duct. (Used with permission from Mafee MF, Valvassori GE, Becker M. *Imaging of the Head and Neck*, 2nd ed. Stuttgart/New York: Thieme;2005:837.)

Surgical

Thyroglossal duct cysts are excised through a transverse incision at the level of the hyoid bone. The center of the hyoid bone is removed to include a tract entering the base of the tongue (Sistrunk's procedure) to minimize recurrence. Avoiding lateral dissection will minimize the possibility of injury to the hypoglossal nerves.

Dermoid cysts and lipomas are excised through a simple transverse neck incision positioned in accordance with the location of the mass.

◆ Complications

Bleeding or hematoma can be prevented using electrocautery on small blood vessels and silk ties on larger vessels. Damage to superior laryngeal nerves or the hypoglossal nerve is also possible. Incomplete excision of the thyroglossal duct increases the risk of recurrence.

◆ Outcome and Follow-Up

The incision should be drained with a Penrose or suction drain. Specimen should be sent to pathology to rule out neoplasm. Prophylactic antibiotics should be given if the excised cyst was infected.

8.17 Congenital Midline Nasal Masses

◆ Key Features

- The most common congenital midline nasal masses are gliomas, encephaloceles, and dermoids.
- The most common of these is the dermoid, derived from ectodermal embryonic tissue.
- A glioma consists of central nervous system (CNS) tissue at the nasal dorsum; it is congenital and not a true neoplasm.
- Meningocele and encephalocele do communicate with the ventricles and contain cerebrospinal fluid (CSF).
- Teratomas are true neoplasms arising from totipotent cells; half of all head and neck teratomas occur in the nose.

Gliomas, meningocele, encephalocele, dermoids, and teratomas exist in the differential diagnosis of the congenital midline nasal mass. A dermoid cyst is a congenital benign neoplasm containing keratinizing squamous epithelium and adnexal skin structures, may be located anywhere from the columella to the nasion, and often presents as a dimple on the nasal bridge. The dermoid may involve only skin and nasal bone, or it may have a true dural connection. Nasal gliomas lack a direct connection to the CNS and contain heterotopic glial elements in a fibrillar stroma, of varying growth rate. Encephalocele is continuous with the CNS. Sincipital encephalocele is present in the nasal dorsum or forehead region, basal encephalocele in the nasopharynx. Encephalocele may include meningocele (containing only meninges), encephalomeningocele (containing brain also), and encephalomeningocystocele (including part of the ventricular system). Midline nasal masses should never be biopsied in the outpatient setting or without prior imaging studies.

◆ Epidemiology

Congenital midline nasal masses occur in 1 in 20 to 40,000 live births. Ten percent of dermoids occur in the head or neck, and 10% of these occur in the nose. Encephalocele occurs in 1 in 1,250 to 2,000 live births; 40% have other abnormalities. Occipital encephalocele is most common in North America (75%), although sincipital ones are most common in Southeast Asia.

◆ Clinical

Signs

Dermoids are typically anywhere along the midline of the nose and are firm, noncompressible, and nonpulsatile. A nasal dermoid sinus or cyst presents with a dimple containing a hair follicle. Rarely, the nasal base is

widened. Encephaloceles are soft, bluish, compressible, pulsatile, and may transilluminate. Gliomas are smooth, firm, noncompressible, and nontransilluminating. All may be associated with a cranial defect, although this is most common with an encephalocele.

Symptoms

Masses may be intranasal or extranasal.

Differential Diagnosis

- Nasal obstruction
- Hypertelorism
- Epiphora
- Infection (local or meningitis)

◆ Evaluation

Physical Exam

- Internal and external nasal exam
- There may be a positive Furstenberg test in encephaloceles (mass expands with jugular vein compression).
- Exam for other possible congenital anomalies

See Table 8.11.

Table 8.11 Clinical approach to congenital midline nasal mass

Evaluation • Head and neck exam • Location, size, quality of mass • Evidence of other congenital anomalies
Furstenberg test • Possible meningocele or encephalocele (enlargement of mass with jugular compression)
Imaging • CT and/or MRI
Presumptive diagnosis • Treatment planning; neurosurgical consult if evidence of CSF connection

Abbreviations: CSF, cerebrospinal fluid; CT, computed tomography; MRI, magnetic resonance imaging.

Imaging

Computed tomography (CT) with contrast; if there is any hint that the mass may be continuous with the CNS, magnetic resonance imaging (MRI) is also required. Ossification of the anterior cranial base is variable in children under 5 years.

Pathology

These masses may result from a failure of complete involution of a dural diverticulum that protrudes through the fonticulus nasofrontalis and typically forms the foramen cecum.

- **Dermoid:** A cyst lined with squamous epithelium, along with hair and sebaceous glands
- **Glioma:** Astrocytes, neuroglial fibers, and S-100 positive
- **Encephalocele:** A nonneoplastic mature neuroglial tissue with meninges

◆ Treatment Options

Small superficial dermoids may be removed via a nasal dorsum incision. A sinus should be removed with an elliptic incision. Other dermoids may require an external rhinoplasty approach and, sometimes, a bicoronal flap if there is extension to the cranial base. For large dermoids, calvarial bone may be needed for reconstruction of cranial base defects.

External gliomas also require an elliptic skin incision or external rhinoplasty approach. Internal gliomas are usually lateral and can be removed through a lateral rhinotomy incision. In either case, if a CSF leak is encountered, a bifrontal craniotomy or an endoscopic CSF leak repair is necessary. This may necessitate collaboration with neurosurgical colleagues for repair.

Encephaloceles require neurosurgical collaboration, and likely a bifrontal craniotomy to maximize visualization. Repair of defects requires temporalis or pericranial flaps for a watertight closure. Extracranial repair is similar to that for gliomas.

◆ Complications

A CSF leak may require lumbar drainage and another patch procedure. Local wound infection or meningitis will require antibiotic therapy:

- Antibiotic coverage for meningitis (see also **Chapter 3.2.3**):
 - Cefotaxime, intravenous (IV), every 4 hours or ceftriaxone, IV, every 12 hours plus ampicillin, IV, every 6 hours
- Alternative if there is drug resistance:
 - Vancomycin (child 15 mg/kg, IV, every 6 hours) plus cefotaxime or ceftriaxone plus ampicillin

Incomplete dermoid or glioma excision will lead to recurrence or fistula formation and require a wide local excision with reconstruction.

◆ Outcome and Follow-Up

Typically, patients are observed in a neurologic intensive care unit (ICU) setting, except for minor external excisions. Patients should be monitored for complete healing and infection.

8.18 Choanal Atresia

◆ Key Points

- Choanal atresia occurs in ~ 1 in 5,000 to 8,000 births.
- Bilateral atresia is more common and is associated with other anomalies (e.g., CHARGE association).
- Endoscopic techniques have greatly improved the safety of the transnasal approach.

Choanal atresia is a congenital condition in which one or both of the choanae are replaced with a bony or mixed bony and membranous wall. Unilateral atresia usually presents later in life, while bilateral atresia is detected at birth. Bilateral atresia can often be medically managed with the use of a McGovern nipple while allowing growth in preparation for repair. The most common repair options include the transnasal and transpalatal approaches. The most frequent complication of repair is restenosis.

◆ Epidemiology

Choanal atresia is relatively rare, only occurring in 1 in 5,000 to 8,000 births. It is almost twice as common in females as in males; 65 to 75% of cases are unilateral. The right choana is more commonly affected. Nearly 75% of bilateral atresias are associated with other disorders, including Treacher Collins's syndrome, DiGeorge's sequence, Apert's syndrome, trisomy 18, and CHARGE association.

◆ Clinical

Signs

Bilateral atresia will most often present in newborns with cyanotic spells relieved by crying and worsened with feeding. Unilateral atresia may present later in childhood with unilateral thick nasal secretions similar to sinusitis.

Symptoms

Bilateral disease will present at birth with difficulty breathing. Unilateral disease may present in older children and may be associated with history of nasal congestion.

◆ Differential Diagnosis

Bilateral

- Congenital nasal piriform aperture stenosis
- NOWCA (nasal obstruction without choanal obstruction)
- Midnasal stenosis
- Choanal stenosis

Unilateral

- Sinusitis
- Nasal foreign body
- Nasal septal deviation
- Choanal stenosis

◆ Evaluation

Physical Exam

Examination of the nares will reveal thick mucous secretions. Commonly, the diagnosis is made after failure to pass a soft suction catheter into the nasopharynx. A flexible endoscope can also be used to assess choanal patency.

Imaging

Computed tomography (CT) provides the most reliable method of diagnosis. Axial images show narrowing of the choanal orifice or widening of the posterior vomer. Approximately 30% of atresias will be purely bony in nature, and 70% will be mixed bony and membranous.

Pathology

There are multiple theories, one of which is that the atresia is due to the persistence of the buccopharyngeal membrane.

◆ Treatment Options

Medical

Initially, an oral airway or a McGovern nipple may be used to maintain an airway. Stimulating the infant to cry will also provide temporary relief. Endotracheal intubation is often appropriate.

Surgical

- *Transnasal approach*: Puncture of the atretic plate using a curved trocar or urethral sound, followed by drilling of the vomer, hard palate, and pterygoid plate. This is usually done with the use of a 120° endoscope in the nasopharynx to provide visualization. Blind use of a trocar is to be avoided. Stents may be placed at the completion of the procedure. Advantages include its ability to be performed in infants and its relatively short operative time. Disadvantages include difficulty in preserving mucosal flaps with the use of the drill. Some surgeons advocate the use of laser (CO₂ or Nd-YAG) with the transnasal approach.
- *Transpalatal approach*: Creation of a palatal flap followed by removal of palatine bone, atretic plate, and vomer. Stents may be placed. Advantages include better visualization, the ability to create lining flaps, and reduced duration of stenting. Disadvantages include oronasal fistula and palatal growth disturbance, with risk of crossbite or midface retrusion.
- *Transantral approach*: Primarily used in revision cases. The choana is opened widely to include the maxillary sinus.

- **Transeptal approach:** Primarily used in unilateral atresia. Removal of posterior septum creates a passage to bypass the atresia.
- The endoscopic technique (nasal or retropalatal), with or without powered instrumentation, offers excellent visualization with great ease in removing the bony choanae. Microdébriders may provide clearer operative fields and causing less tissue trauma for experienced surgeons. Carbon dioxide and potassium titanyl phosphate (KTP) lasers are sometimes used. The use of mitomycin C topically as an adjunct to the surgical repair of choanal atresia may offer improved patency results.

◆ Outcome and Follow-Up

Postoperative care should include daily suctioning of stents, serial endoscopy, and dilation. The rate of restenosis varies widely, with reports between 50 and 100%. Both transnasal and transpalatal approaches seem to have similar rates of restenosis.

8.19 Cleft Lip and Palate

◆ Key Features

- Group I: Cleft lip
 - Unilateral or bilateral
 - Complete (extension to nasal floor)
 - Incomplete (muscle diastases: vermillion to bridge of tissue at nasal sill)
- Group II: Cleft palate
 - Unilateral or bilateral
 - Secondary palate only
- Group III: Cleft lip and cleft palate
 - Complete cleft palate (both primary and secondary palate)
- Group IV: Cleft palate
 - Unilateral or bilateral
 - Primary palate only

Cleft lip and cleft palate can be classified as syndromic (15–60%) or nonsyndromic. Syndromic associations include Apert's, Stickler's, Treacher Collins's, and Waardenburg's syndromes. Cleft lip with or without cleft palate is believed to be genetically distinct from isolated cleft palate. Nonsyndromic clefts may be secondary to exposure to teratogens (e.g., ethanol, anticonvulsants, steroids, vitamin A excess). Likewise, maternal and intrauterine factors such as gestational diabetes, smoking, or amniotic bands may play a role.

Typically, cleft palate management is done by a team including surgeons, dentists, orthodontists, speech pathologists, and audiologists.

◆ Embryology of the Lip and Palate

The palate is embryologically divided into the primary (the maxillary alveolus and palate anterior to the incisive foramen), and secondary (originating posterior to the incisive foramen and terminating at the uvularis) components. Fusion of the paired median nasal prominences (MNPs) gives rise to the primary palate. This process initiates the separation of the oral from the nasal cavity. The coalescence of the MNPs forms the central maxillary alveolar arch, central and lateral incisors, anterior hard palate, premaxilla, philtrum of the upper lip, columella, and nasal tip. The genesis of the central upper lip occurs in association with the primary palate. The frequent association of these deformities is not unexpected. The remainder of the upper lip, lateral to the philtrum, is formed by fusion of the paired MNPs medially with the maxillary processes laterally.

Formation of the secondary palate is initiated by contact of the nasal septum to the lateral palatal shelves at the incisive foramen. The medialization of the palatal shelves then occurs in an anterior to posterior direction. The midline tongue represents a barrier to this medialization. Development of the mandible results in anterior displacement of the tongue, allowing normal growth of the palatal shelves. Failure of this process results in the Pierre Robin's sequence (micrognathia, relative macroglossia, and U-shaped cleft palate); see also **Chapter 8.8**.

There is not only an embryologic structural division of the palate, but a temporal one as well. The primary palate forms during weeks 4 to 7 of gestation. The secondary palate begins development after completion of the primary palate and occurs during weeks 8 to 12 of gestation.

◆ Epidemiology

Cleft lip and palate are the most common congenital malformations of the head and neck. The incidence of cleft lip with or without cleft palate in the United States is 1:1,000 newborns and varies according to race, with the highest incidence in Native Americans and a male/female ratio of 2:1. The incidence of cleft palate is 1:2,000 and is equal across ethnic groups, with a male/female ratio of 1:2.

◆ Clinical

Signs

See the classification of defects noted under Key Features. Signs of submucous cleft palate include:

- Bifid uvula
- Zona pellucida
- Notched hard palate

Symptoms

The symptoms are feeding difficulties with nasal regurgitation.

Differential Diagnosis

Associated syndromes include:

- Pierre Robin's sequence: Micrognathia, glossoptosis, and U-shaped cleft palate
- Stickler's syndrome: Retinal detachment, cataracts, and early arthritis
- Treacher Collins's syndrome: Eyelid colobomas, middle ear ossicular abnormalities, and malformation of facial bones
- Apert's syndrome: Acrocephaly, fused digits, and stapes fixation

◆ Evaluation

Physical Exam

- Determine type of defect:
 - Unilateral, bilateral, median
 - Complete (extension to nasal floor) or incomplete (submucosal)
 - Primary (anterior to incisive foramen) or secondary (posterior to incisive foramen)
- Look for associated defects:
 - Facial defects: telecanthus, maxillary/malar hypoplasia, nasal deformities, facial nerve paralysis
 - Otologic anomalies, which should be examined if they exist
 - Synostoses
- Determine presence of associated syndrome. Apert's, Stickler's, Treacher Collins's, Waardenburg's syndromes, and Pierre Robin's sequence are associated with cleft palate.
- Cleft lip nasal deformity

Dehiscence of the orbicularis oris muscle results in its abnormal, nonanatomic insertion laterally onto the ala and medially onto the columella. The subsequent muscular tension produces a characteristic nasal deformity. The nasal tip and columella are deflected to the noncleft side. The cleft-side ala is positioned laterally, inferiorly, and posteriorly. The cleft nostril is widened and horizontally oriented. The septum is bidirectionally affected. Whereas the caudal aspect deflects toward the noncleft side, the remainder of the cartilaginous and bony septum deflects toward the cleft side. This results in decreased air entry through both nasal passages.

The bilateral cleft nasal deformity is dependent on the severity of the individual sides. If both sides are equally involved, the nasal tip is typically midline, poorly defined, and frequently bifid. Both sides are composed of obtuse domal angles and widened, horizontally oriented nostrils. The alae are both positioned laterally, inferiorly and posteriorly. The septum is midline. If there is asymmetry of the cleft lip, deformities of the nasal tip, columella, and septum are deviated to the less affected side; however, the deflection is less apparent than it would be with a normal side.

Imaging

Adjunctive imaging may be required for the investigation of syndromic etiology.

Other Tests

Close audiologic monitoring is warranted given the increased incidence of acute otitis media (OM) and serous OM due to eustachian tube dysfunction in this population.

◆ Treatment Options

Medical

- Parental counseling
- Ensure adequate feeding and nutrition
 - Cleft lip and alveolus—often feed normally by bottle or breast
 - Complete cleft lip or palate—often have feeding problems initially
 - Inability to generate sufficient seal around the nipple
 - Increased work of feeding and swallowing of air
 - Specialty nipple may be required
 - Premie nipple
 - Haberman feeder
 - Mead-Johnson cross-cut nipple
 - Squeeze bottle nurser
 - Palatal prosthesis for wide clefts for continued feeding difficulties

Surgical

- Lip adhesion
 - Converts a complete cleft lip into an incomplete cleft lip
 - Performed at 2 to 4 weeks of age
 - Allows definitive lip repair to be performed under less tension at 4 to 6 months of age
 - Criteria:
 - Wide unilateral complete cleft lip and palate where conventional lip repair would produce excessive incisional tension
 - Symmetric wide bilateral complete cleft lip with prominent premaxilla
 - Converts an asymmetric bilateral cleft lip to a symmetric cleft lip
 - Disadvantage: Scar tissue that may later interfere with definitive surgical management of the lip
- Cleft lip repair
 - “Rule of tens” for timing of cleft lip repair
 - Age ≥ 10 weeks
 - Weight ≥ 10 lb
 - Hemoglobin ≥ 10 g

- Surgical approaches:
 - Millard rotation advancement technique (**Fig. 8.6**)
 - Tennison-Randall: Single triangular flap interdigitation (**Fig. 8.7**)
 - Bardach: Double triangular flap interdigitation
 - Straight line closure (uncommon)

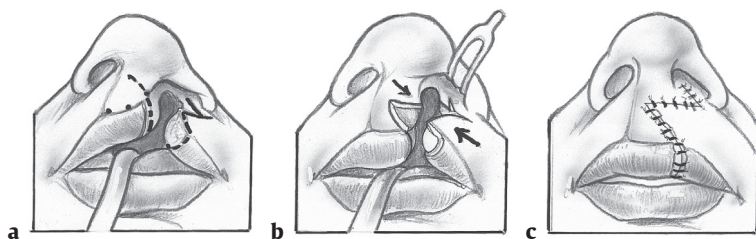


Fig. 8.6 (a,b,c) Millard rotation advancement lip repair.

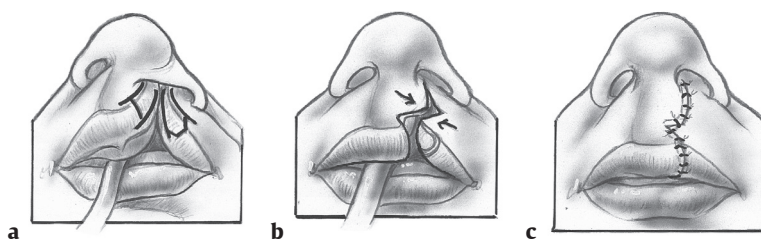


Fig. 8.7 (a,b,c) Tennison-Randall triangular flap lip repair.

- Cleft palate repair
 - The exact timing of surgical closure of cleft palate is controversial.
 - There is a balance between establishing the velopharyngeal competence necessary for speech and the potential negative influence of early repair on maxillofacial growth and occlusion.
 - It is usually performed between 9 and 16 months of age.
 - Surgical approaches:
 - Schweckendiek: Closure of soft palate only
 - Von Langenbeck: Two bipedicle mucoperiosteal flaps are created by incising along the medial cleft edges and along the posterior alveolar ridge from the maxillary tuberosities to the anterior level of the cleft. They are mobilized medially and closed in layers (**Fig. 8.8**).
 - Bardach: Two-flap palatoplasty for complete cleft palate repair
 - Furlow palatoplasty: Double Z-plasty for secondary cleft palate repair (**Fig. 8.9**)
 - V-Y pushback technique for secondary cleft palate repair
 - Tympanostomy tubes are commonly inserted at the time of palate repair.

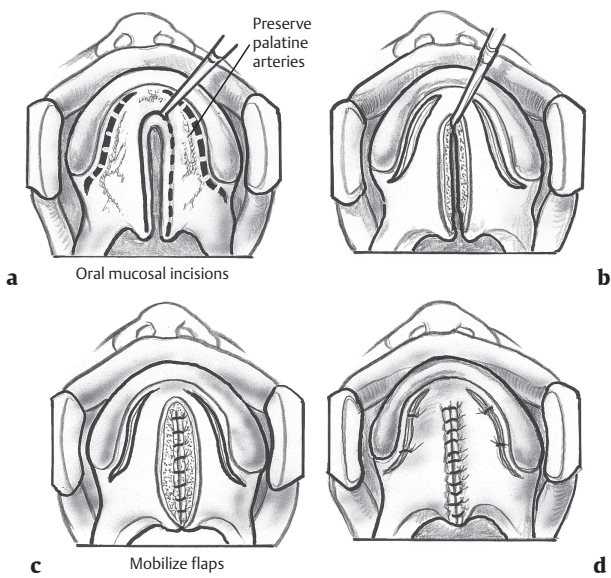


Fig. 8.8 (a–d) Von Langenbeck repair.

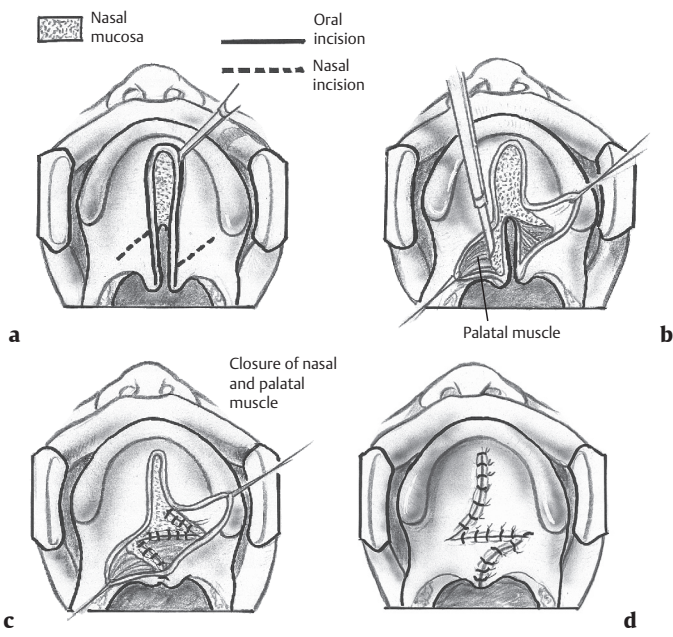


Fig. 8.9 (a–d) Furlow palate lengthening procedure ("double opposing Z-plasty").

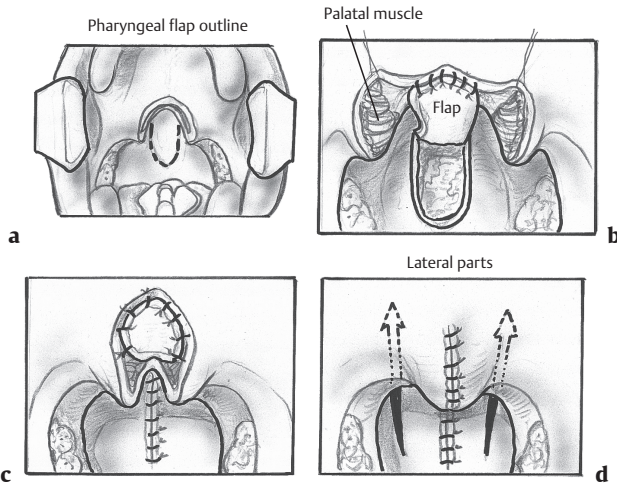


Fig. 8.10 (a–d) Palatal flap.

◆ Complications

Primary or secondary hemorrhage is uncommon but may require a return to the OR for management. Oronasal fistula is managed with secondary repair. Velopharyngeal insufficiency is failure of closure of the velopharyngeal sphincter, resulting in incomplete separation of the nasal cavity from the oral cavity. During speech, air leaks into the nasal cavity, resulting in hypernasal vocal resonance and nasal emissions. During feeding, it can result in nasal regurgitation of food. Treatment options may include speech therapy, palate lengthening (**Fig. 8.9**), a palatal flap (**Fig. 8.10**), and augmentation of the posterior pharyngeal wall.

◆ Outcome and Follow-Up

Postoperative care should include pain control, the establishment of a feeding plan, and monitoring of the airway. Routine surgical follow-up is required for cleft repairs at 4 weeks with growth and feeding assessment. Ongoing evaluation and management of the middle ear in combination with audiologic assessment should be completed every 6 months. There should be a referral to a speech therapist for assessment and therapy.